

Category	Subsystem / Focus	Sheet Link	Description / Quick Notes
0.0 Program Control	Mission Overview	M.S. & Objectives	High-level milestones & project goals.
	Systems Engineering	Requirements	Master system and subsystem tracking.
1.0 Payload & Science	Core Instruments	Payload	Breakdown of scientific sensors and components.
2.0 Operations	Mission Timeline	Outline Key Phases	Chronological breakdown of mission elapsed time.
		Concept of Operations Timeline	
		minute-by-minute breakdown	
3.0 Mechanical & Structure	Mass Tracking	Mass Budget	Margin calculations and weight allocations.
	Hardware Trade Studies	Structures Decision Matrix	Comparative analysis of chassis designs.

Level		OBJECTIVES & REQUIREMENTS	Objective or Requirement	Date Last Updated	Source(s)	System/ Subsystem	Status?	VERIFICATION & VALIDATION	
OBJ = Objective, REQ = Requirement, Constraint. Hierarchy = 01.02.03 (01 = Objective Parent Category, 02 = First Child Requirement, 03 = Second Child Requirement... etc)			Full statement sentence. Objectives are the requirements in simple, easy to understand, general terms. Requirements are technical as "shall" statements with a specific, measureable, and testable guideline to be met.	Date of when the requirement statement was last revised.	Where did this requirement come from? Is this based on literature, stakeholder, testing standards, etc? Include links if possible.	Does this matter on a system level? If not, which subsystems does this directly affect? Consider the interfaces as well. Which subsystems are responsible?	Accomplished or unaccomplished?		
MISSION STATEMENT	MISSION STATEMENT		The multi-agent Europa surface system shall autonomously characterize the ice shell and identify a cryobot insertion site, operating without real-time Earth support.	6/14/2026	-	-	-		
OBJ.01	System Level		The system shall demonstrate autonomous multi-agent coordination and consensus-based decision making without Earth communication links	6/14/2026	This objective addresses the multi-agent coordination shortfall ranked 139th in NASA's Civil Space Technology Shortfall Rankings, which identifies cooperative task planning as a near-term priority (STMD) This objective addresses NASA's identified gap in autonomous navigation without Earth-based support, as stated in the FY 2026-2027 SBIR solicitation Topic COMNAV.1.S26B [SBIR] This objective supports the science goals outlined in the Decadal Survey, which identifies Europa's subsurface ocean as a primary target for astrobiology and calls for characterization of the ice shell to assess habitability. [OWL] This objective addresses NASA's identified gap in autonomous distributed spacecraft operations without Earth-based navigational support, per SBIR Topic COMNAV.1.S26B. [SBIR]				
OBJ.02	Subsystem Mobility		The rover shall traverse Europa's surface terrain and maintain localization using onboard sensors and inter-agent ranging	6/14/2026					
OBJ.03	Subsystem Science/Payload		The system shall characterize ice shell thickness using a ground penetrating radar sounder and identify candidate insertion sites	6/14/2026					
OBJ.04	Subsystem COMMs		The inter-agent mesh network shall maintain coordination under degraded connectivity and radiation induced faults	6/14/2026					

Testing Method	Verification	Validation	Any Pre-
----------------	--------------	------------	----------

	<i>Verification - Does it meet the specification need? Are we building our product right?</i>	<i>Validation - Does it meet the customer need? Did we build the right product?</i>	
--	---	---	--

Level		OBJECTIVES & REQUIREMENTS	Objective or Requirement	Date Last	Reasoning	Source(s)	System/
OBJ = Objective, REQ = Requirement, CSR = Constraint. Hierarchy = 01.02.03 (01 = Objective Parent Category, 02 = First Child Requirement, 03 = Second Child Requirement... etc)			Full statement sentence. Objectives are the requirements in simple, easy to understand, general terms. Requirements are technical as "shall" statements with a specific, measureable, and testable guideline to be met.	Date of when the requirement statement was last revised.		Where did this requirement come from? Is this based on literature, stakeholder, testing standards, etc? Include links if possible.	Does this matter on a system level? If not, which subsystems does this directly affect? Consider the interfaces as well. Which subsystems are responsible?
REQ.01.01	System [Mission level]		The multi-agent Europa system shall operate fully autonomously for the entire surface mission duration without requiring real-time command uplinks from Earth.	6/14/2026	The 44-minute one-way light delay between Earth and Jupiter makes real-time demanding physically impossible, all decisions must be onboard. Traceable to OBJ.01	NASA FY 2026-2027 SBIR Solicitation, Topic COMNAV.1.S26B, 2025	System
REQ.01.02	System [Mission level]		The system shall survive and maintain operational status after exposure to total ionizing dose of no less than 100 krad over the nominal mission lifetime.	6/14/2026	Jupiter's magnetosphere delivers radiation doses far exceeding near-Earth missions. Europas Clipper's radiation vault architecture targets ~100 krad as the design threshold. Radiation environment directly drives the shielded vault requirement. Traceable to OBJ.01	National Academies, Origins Worlds and Life Decadal Survey 2023–2032, https://www.nationalacademies.org/read/26522 ; Galileo spacecraft radiation vault design heritage	System, Electronics, Power
REQ.01.03	System [Mission level]		The system shall maintain a minimum inter-agent network connectivity of two active relay nodes at all times during surface operations to ensure consensus is achievable.	6/14/2026	Consensus algorithms require a minimum connected graph to guarantee convergence. Two active relays plus a surface rover guarantee a 3-node connected topology as the minimum viable configuration. Traceable to OBJ.01 and OBJ.04.	NASA STMD Civil Space Technology Shortfall Rankings, Jul. 2024, https://www.nasa.gov/wp-content/uploads/2024/07/civil-space-shortfall-ranking-july-2024.pdf	System, COMMs
REQ.02.01	Subsystem [Operations Level] Mobility		The surface robot shall maintain a localization accuracy of no worse than +-0.5 m relative to its last known position during any traverse segments up to 50 m in the absence of GPS.	6/14/2026	Cryobot insertion site selection requires meter-level positioning to match the radar identified thin ice candidate. +-0.5 over 59 m segments in consistent with visual odometry performance on Mars rovers. Traceable to OBJ.02	NASA FY 2026-2027 SBIR Solicitation, Topic COMNAV.1.S26B; Mars Curiosity rover visual odometry heritage	Mobility, GNC
REQ.03.01	Subsystem [Operations Level] Science / Payload		The ground-penetrating radar sounder shall resolve ice shell thickness to within +-10% accuracy for ice layers between 1 km and 30 km depth, producing a subsurface map at the minimum spatial resolution of 100 m along the traverse track.	6/14/2026	Identifying optimal cryobot insertion site requires knowing where the ice shell is the thinnest. +- 10% thickness accuracy is enough to distinguish viable drilling candidates. The 100 m along track resolution is consistent with the Europa Clipper REASON instrument design parameters. Traceable to OBJ.03	National Academies, Origins Worlds and Life Decadal Survey 2023–2032, https://www.nationalacademies.org/read/26522; Europa Clipper REASON instrument documentation	Science/Payload, Power
REQ.04.01	Subsystems [Operations Level] COMMS		The inter-agent mesh network shall achieve consensus on a shared terrain map update within no more than 10 minutes under a degraded network condition defined as packet loss rates of up to 30% on any single inter-agent link	6/14/2026	Rover traverse planning depends on a current consensus validated terrain map. A 10 minute convergence bound limits traverse planning delay to an acceptable operations cadence. 30% packet loss represents a plausible radiation induced degradation scenario for the environment. Traceable to OBJ.04.	NASA STMD Civil Space Technology Shortfall Rankings, Jul. 2024; NASA FY 2026-2027 SBIR Solicitation, Topic COMNAV.1.S26B	COMMs, CDH

Level		OBJECTIVES & REQUIREMENTS	Objective or Requirement	Date Last	Reasoning	Source(s)	System/
OBJ = Objective, REQ = Requirement, Constraint. CSR = Number Hierarchy = 01.02.03 (01 = Objective Parent Category, 02 = First Child Requirement, 03 = Second Child Requirement... etc) System, Subsystem, or Component Level?			Full statement sentence. Objectives are the requirements in simple, easy to understand, general terms. Requirements are technical as "shall" statements with a specific, measurable, and testable guideline to be met.	Date of when the requirement statement was last revised.	Where did this requirement come from? Is this based on literature, stakeholder, testing standards, etc? Include links if possible.	Does this matter on a system level? If not, which subsystems does this directly affect? Consider the interfaces as well. Which subsystems are responsible?	
REQ.03.02	Subsystem [Operation Level] Science / payload		The radar sounder antenna shall be mounted such that the antenna boresight remains within +-5 of nadir during all active sounding operations; the rover shall halt traversal if the measured surface slope exceeds 20 to maintain pointing compliance.	6/16/2026	A nadir pointing antenna is required to minimize geometric error in the two-way travel time measurement. . Off nadir tilt reduces basal reflection return strength and introduces range error that degrades	National Academies, Origins Worlds and Life Decadal Survey 2023–2032; Europa Clipper REASON instrument documentation	Science/Payload, Structures, Mobility
REQ.03.03	Subsystem [Operations Level] Science / Payload		The rover power subsystem shall reserve a dedicated peak power allocation for radar sounder transmission pulses and shall command drive motor shutdown during each active sounding window to prevent power contention with the RTG continuous output.	6/16/2026	Radar sounders draw high peak pulse power during transmission. The RTG provides fixed continuous output; simultaneous drive and sounding would exceed available power margin and risk under-powering the radar pulse, degrading return signal amplitude below the noise floor. Stop-and-sound is consistent with high-power instrument operation heritage on Mars rovers. Traceable to OBJ.03. At approximately 40 traces per sol across four data sheets, raw science data accumulates faster than the non-deterministic Earth relay can downlink. Seven sols of buffer covers the consensus convergence window and plausible communication gaps. Lossless compression preserves the measurement fidelity required by REQ.03.01 while reducing relay load. Traceable to OBJ.03 and OBJ.04.	Europa Clipper REASON instrument documentation; NASA RTG power system heritage	Science/Payload, Power, Mobility
REQ.03.04	Subsystem [Operations Level] Science / Payload		The command and data handling subsystem shall provide onboard storage sufficient to retain a minimum of 7 sols of uncompressed radar trace data, and shall apply lossless compression to all Radar_Traces records prior to inter-agent relay transmission, achieving a minimum compression ratio of 2:1.	6/16/2026	Europa's surface temperature is approximately 90K (-183C). Battery cells experience severe capacity loss below -20C and permanent damage below -40C. The RTG provides continuous waste heat that can be routed to a thermal enclosure, making active heating feasible without a dedicated power draw penalty. No ground intervention is possible to recover a thermally failed rover. Traceable to OBJ.01.	NASA STMD Civil Space Technology Shortfall Rankings, Jul. 2024; Europa Clipper REASON instrument documentation	Science/Payload, COMMs, CDH
REQ.05.01	Subsystem [Operational Level] Thermal		The rover thermal control system shall maintain all avionics and battery cells within an operational temperature range of 0C to 40C during all active surface operations and above -20C during night-mode low-power thermal management, using active heater elements powered by RTG waste heat.	6/28/2026		National Academies, Origins Worlds and Life Decadal Survey 2023–2032; NASA RTG thermal heritage	Thermal, Power, Structures

Level		OBJECTIVES & REQUIREMENTS	Objective or Requirement	Date Last	Reasoning	Source(s)	System/
OBJ = Objective, REQ = Requirement, Constraint. CSR = Number Hierarchy = 01.02.03 (01 = Objective Parent Category, 02 = First Child Requirement, 03 = Second Child Requirement... etc) System, Subsystem, or Component Level?			Full statement sentence. Objectives are the requirements in simple, easy to understand, general terms. Requirements are technical as "shall" statements with a specific, measurable, and testable guideline to be met.	Date of when the requirement statement was last revised.		Where did this requirement come from? Is this based on literature, stakeholder, testing standards, etc? Include links if possible.	Does this matter on a system level? If not, which subsystems does this directly affect? Consider the interfaces as well. Which subsystems are responsible?
REQ.05.02	Subsystem [Operations Level] CDH		The command and data handling subsystem shall detect and autonomously recover from a single-event latchup in any processor core within 330 seconds of occurrence, restoring nominal operations without ground intervention, and shall log each latchup event to non-volatile memory with a timestamp and affected core identifier.	6/28/2026	Jupiter's magnetosphere produces a high-energy particle flux that induces single event latchup at a far higher rate than near-Earth missions. With a 44-minute one-way light delay, no ground intervention can recover a locked core in time to prevent missions tall. The 30-second recovery bound prevents autonomous traverse planning and site selection from stalling indefinitely. Traceable to OBJ.01, linked to RISK-02.	National Academies, Origins Worlds and Life Decadal Survey 2023–2032; NASA STMD Civil Space Technology Shortfall Rankings, Jul. 2024; Galileo spacecraft radiation heritage	CDH, Electronics
REQ.05.03	Subsystem [Operations Level] Power		The rover power management subsystem shall enforce a per-sol energy budget such that total energy draw across all subsystems does not exceed 90% of the RTG's nominal daily energy output, reserving a minimum 10% margin for autonomous anomaly response, heater activations, and unplanned communication relay passes.	6/28/2026	The RTG provides fixed continuous output with no ability to surge on demand. Radiation-induced matchups, slope-stop halts per REQ.03.02, and degraded consensus requiring extra relay passes all introduce unscheduled power draws. A hard 10% reserve prevents the rover from entering a power fault state during autonomous recovery events with no ground support available. Traceable to OBJ.01 and OBJ.03	NASA RTG power system heritage; Europa Clipper power budget architecture; NASA FY 2026–2027 SBIR Solicitation, Topic COMNAV.1.S26B	Power, CDH, Thermal, Mobility
REQ.06.01	Subsystem [Operation Level] Mobility		The rover shall autonomously plan and execute a surface traverse of no less than 100m per sol across Europa analog terrain, pausing at intervals no greater than 100m along-track to perform a radar sounding stop, and shall resume traversal upon sounding completion without requiring a ground uplink.	6/28/2026	The mission objective requires building a spatially continuous subsurface ice map at <100m along-track resolution (REQ.03.01). To meet that resolution the rover must stop and sound at least every 100m, then continue. With no real-time Earth support (REQ.01.01), the stop-and-sound cycle must be fully onboard. The 100m/sol minimum traverse rate ensures the science map grows at a pace consistent with the 30-sol surface operations window. Traceable to OBJ.02 and OBJ.03.	National Academies, Origins Worlds and Life Decadal Survey 2023–2032; Europa Clipper REASON instrument documentation; NASA FY 2026–2027 SBIR Solicitation, Topic COMNAV.1.S26B	Mobility, GNC, Science/Payload
REQ.06.02	Subsystem [Operations Level] Science / Payload		The rover's onboard site-selection algorithm shall autonomously evaluate each NOMINAL-flagged radar trace against the cryobot insertion criteria (ice thickness < 5 km, surface slope < 20. terrain roughness index <0.7) and shall flag a candidate insertion site within one sol of the criteria first being satisfied without a ground intervention.	6/28/2026	The terminal mission objective is identifying an optimal cryobot insertion site autonomously. With a 44-minute light delay, Earth cannot evaluate radar data and issue a site selection decision in real time. The onboard algorithm must continuously evaluate incoming traces and trigger the site flags as soon as conditions are met. The one-sol response bound ensures the mission does not waste traverse budget driving past a viable site. Traceable to OBJ.03.	National Academies, Origins Worlds and Life Decadal Survey 2023–2032; Europa Clipper REASON instrument documentation; NASA FY 2026–2027 SBIR Solicitation, Topic COMNAV.1.S26B	Science/Payload, CDH

Level		OBJECTIVES & REQUIREMENTS	Objective or Requirement	Date Last	Reasoning	Source(s)	System/
OBJ = Objective, REQ = Requirement, Constraint. CSR = Number System, Subsystem, or Component Level? Hierarchy = 01.02.03 (01 = Objective Parent Category, 02 = First Child Requirement, 03 = Second Child Requirement... etc)			Full statement sentence. Objectives are the requirements in simple, easy to understand, general terms. Requirements are technical as "shall" statements with a specific, measurable, and testable guideline to be met.	Date of when the requirement statement was last revised.		Where did this requirement come from? Is this based on literature, stakeholder, testing standards, etc? Include links if possible.	Does this matter on a system level? If not, which subsystems does this directly affect? Consider the interfaces as well. Which subsystems are responsible?
REQ.06.03	Subsystem [Operations Level] COMMs		The rover shall autonomously transmit a daily science summary packet to the orbital relay network containing the sol's Daily Summary record (traverse distance, valid trace count, minimum ice thickness, candidate site count, and mean consensus convergence time) within 2 hours of completing each sol's radar data collection window, without requiring a ground-initiated downlink command/	6/28/2026	Earth-based scientists and mission planners require timely sol-by-sol progress data to assess whether mission objectives are being met and to prepare contingency plans within the non-deterministic relay window. With a 44-minute light delay, the rover cannot wait for a downlink command; it must autonomously initiate transmission when the data is ready. The 2-hour bound ensures the daily summary reaches before the next sol's traverse planning begins, maintaining operational cadence. Traceable to OBJ.01 and OBJ.04	NASA STMD Civil Space Technology Shortfall Rankings, Jul. 2024; NASA FY 2026–2027 SBIR Solicitation, Topic COMNAV.1.S26B	COMMS, CDH, Science/Payload
REQ.07.01	Subsystem [Design] Structures		The rover chassis and landing leg structure shall accommodate vertical landing loads of 12 g and lateral loads of 6 g touchdown, with each leg sized to distribute peak contact forces across four footpads without permanent deformation or buckling.	7/7/2026	Europa Clipper impact analysis shows worse-case terrain encounters. Landing dynamics on low- gravity icy surface require robust leg geometry to prevent tipping or frame cracking. Traceable to mission landing phase success.	Nacional Academies Decadal Survey: Mars rover landing heritage (Perseverance suspension); Europa Clipper mission architecture	Structures, Mobility, GNC
REQ.07.02	Subsystem [Design] Structures		The primary equipment deck, fastener selection, and all structural material shall remain within material specification limits (yield strength, fatigue life) when exposed to sustained cryogenic operation at EUropa surface temperature of 90k (-183c) over the 30- sol mission lifetime.	7/7/2026	Aluminum and titanium alloys experience reduced ductility and altered fatigue properties at cryogenic temperatures. Fastener materials must remain ductile to avoid brittle fracture during thermal cycling. Noground repair possible. Traceable to OBJ.01	National Academies Decadal Survey: Galileo spacecraft cryogenic experience; MMPTE cryogenic materials database	Structures, Thermal, Power

Level	OBJECTIVES & REQUIREMENTS	Objective or Requirement	Date Last	Reasoning	Source(s)	System/
OBJ = Objective, REQ = Requirement, Constraint. CSR = Number System, Subsystem, or Component Level? Hierarchy = 01.02.03 (01 = Objective Parent Category, 02 = First Child Requirement, 03 = Second Child Requirement... etc)		Full statement sentence. Objectives are the requirements in simple, easy to understand, general terms. Requirements are technical as "shall" statements with a specific, measureable, and testable guideline to be met.	Date of when the requirement statement was last revised.		Where did this requirement come from? Is this based on literature, stakeholder, testing standards, etc? Include links if possible.	Does this matter on a system level? If not, which subsystems does this directly affect? Consider the interfaces as well. Which subsystems are responsible?
REQ.07.03		The radiation shielding valut enclosing the flight computer and power distribution unit shall provide sufficient material thickness to attenuate the Europa radiation environment to0 a dose rate of <= 1 krad per sol at the electronics level, maintaining structural stiffness and mass allocation within the 28.5 kg structures budget.	7/7/2026	Radiation dose accumulation (REQ. 01.02) is mitigated by shielding geometry and material selection. nRTG and shielder vault approach (Europa Clipper heritage) required structural integration of shielding layers. Balances protection vs mass. Traceable to REQ.01.02, OBj.01.	Europa Clipper radiation vault design; Galileo spacecraft documentation; JPL REASON instrument shielding analysis	Structures, Power, Electronics, CDH
REQ.08.01		The power system shall generate a minimum of 110 We at the beginning of life via an MMRTG--class radioisotope generator and shall deliver no less than 80 We continuous electrical output upon European arrival, providing at least 1.9 kWh of usable energy per sol throughout the 30-sol surface The energy storage system shall provide a minimum of 2.4 kWh usable capacity to 1) load-level intermittent peak draws (drive motors, radar transmission, X-band downlink) exceeding continuous RTG output, and 2) sustain night-mode survival heating and critical avionics for no less than 12 hours. while	7/13/2026	Solar power is not viable at Jupiter distance. MMRTG output degrades ~4.8% per year so a110 We BOL unit delivers ~82 We after the 6-year cruise. Sizing to end-of-cruise output rather than BOL ensures the steady-state surface loads (avionics, heaters, commons) is covered with margin for the full The RTG provides fixed continuous output with no surge capability; peak surface loads (~100-125 W during traverse and sounding) exceed the ~80 We available, so the battery must buffer the difference. Cell capacity drops sharply at cryogenic temperatures, driving the co-located thermal requirement. The 344-minute-one-way light delay makes ground-commanded fault recovery impossible, and Jupiter's radiation environment makes lathrup-induced overcurrent events likely (REQ.05.02). A cascading bus fault during night mode would be mission-terminal. The PDU is housed inside the radiation vault per REQ.07.04, but fault containment logic must still bound any single fault. Traceable to OBJ.01, Req. 05.02, and REQ.05.03	Perseverance/Curiosity MMRTG performance heritage; NASA RTG power system heritage; Europa Clipper power architecture	Power, Thermal, System
REQ.08.02		Subsystem [EPS, Battery Storage and Thermal Control]	7/13/2026		Mass budget battery sizing (19.2 kg, 150 Wh/kg); NASA primary battery design heritage; National Academies OWL Decadal Survey 2023–2032	Power, Thermal
REQ.08.03		Subsystem [EPS,Power distribution, and Fault Management]	7/13/2026		NASA Fault Management Handbook; Galileo spacecraft radiation heritage; Europa Clipper radiation vault design	Power, CDH, Electronics

Level		OBJECTIVES & REQUIREMENTS	Objective or Requirement	Date Last	Reasoning	Source(s)	System/
OBJ = Objective, REQ = Requirement, Constraint. CSR = Number Hierarchy = 01.02.03 (01 = Objective Parent Category, 02 = First Child Requirement, 03 = Second Child Requirement... etc) System, Subsystem, or Component Level?			Full statement sentence. Objectives are the requirements in simple, easy to understand, general terms. Requirements are technical as "shall" statements with a specific, measureable, and testable guideline to be met.	Date of when the requirement statement was last revised.	Where did this requirement come from? Is this based on literature, stakeholder, testing standards, etc? Include links if possible.	Does this matter on a system level? If not, which subsystems does this directly affect? Consider the interfaces as well. Which subsystems are responsible?	
REQ.09.01	System [Thermal Control, Heat Rejection]		The rover thermal control system shall reject up to 500 W of combined electronics dissipation and routed RTG waste heat through dedicated radiator panels to maintain the avionics vault interior below +40 C during peak daytime radar-sounding and traverse operations.	7/20/2026	Radar transmit electronics, drive motors, and comms transmitters dissipate significant heat during active operations; without dedicated rejection the sealed vault exceeds the +40 C avionics limit set in REQ. 05.01. Radiators sized for 400-500 W (mass budget) provide the hot-case margin.	NASA MMRTG waste-heat thermal heritage; rover mass budget radiator sizing (2 panels, 400-500 W rejection @ 6 kg/m^2); NASA DSNE thermal design standards	Thermal, Electronics, Power, CDH
REQ.09.02	Subsystem [Operational Level] Thermal / Mobility		The thermal control system shall preheat all drive-train actuators and deployment mechanisms to above -40 C prior to any commanded motion, completing cold-start warm-up in no more than 60 minutes from the 90 K surface baseline to prevent lubricant stiffening and mechanical binding.	7/20/2026	At 90 K, actuator lubricants stiffen and mechanisms risk binding or fracture if driven cold. A bounded pre-motion warm-up protects the drive train and matches the 60-minute Systems Warm-Up block already defined in the ConOps.	Mars Curiosity/Perseverance actuator cold-start and warm-up heritage; NASA DSNE design standards; Europa surface temperature ~90 K, OWL Decadal Survey 2023-2032	Thermal, Mobility, Structures
REQ.09.03	Subsystem [Thermal Control, Passive Architecture]		The thermal control system shall combine multi-layer insulation with routed RTG waste heat to bound the equipment-enclosure heat leak at the 90 K environment such that supplemental electric survival heating does not exceed 50 W and remains within the per-sol energy budget defined in REQ.05.03.	7/20/2026	RTG waste heat (~1.9 kW thermal) is the primary survival heat source; MLI limits radiative loss so electric heaters stay minimal, keeping night thermal draw inside the per-sol energy budget. This trade is the subject of the night-survival thermal trade study.	NASA MMRTG waste-heat thermal heritage; Galileo / Europa Clipper MLI thermal architecture; rover power budget (2006.5 Wh/sol, REQ. 05.03)	Thermal, Power, Structures
REQ.10.01	Subsystem [Operations Level] COMMs		The communications subsystem shall deliver no less than 55 MB of compressed science and engineering data per Earth day from the surface rover to the orbital relay network, sustaining a return-link data rate of at least 128 kbps and accepting a forward-link command rate of at least 8 kbps.	7/28/2026	The rover generates ~90 MB of raw data per sol across radar sounder traces, nav camera stereo imagery, localization logs, terrain and consensus map deltas, and housekeeping. At the 2:1 compression already required by REQ.03.04 that is 45 MB, and a 20% margin gives 55 MB. Clearing 55 MB inside the 90-minute relay pass the ConOps allocates needs 90 kbps, so 128 kbps is the floor at the next standard step up. Forward traffic is only sequence loads and occasional software patches, so 8 kbps is sufficient.	Rover data product definition (2.1 Radar Traces / Localization Log / Terrain and Health sheets); REQ. 03.04 onboard storage and 2:1 compression; ConOps relay pass allocation (90 min/sol); JPL Electra / Electra-Lite UHF proximity link heritage (MRO, MSL, Perseverance); CCSDS 211.1-B Proximity-1	COMMs, CDH, Science/Payload, Power

Level	OBJECTIVES & REQUIREMENTS	Objective or Requirement	Date Last	Reasoning	Source(s)	System/
OBJ = Objective, REQ = Requirement, Constraint. CSR = Number System, Subsystem, or Component Level? Hierarchy = 01.02.03 (01 = Objective Parent Category, 02 = First Child Requirement, 03 = Second Child Requirement... etc)		Full statement sentence. Objectives are the requirements in simple, easy to understand, general terms. Requirements are technical as "shall" statements with a specific, measureable, and testable guideline to be met.	Date of when the requirement statement was last revised.		Where did this requirement come from? Is this based on literature, stakeholder, testing standards, etc? Include links if possible.	Does this matter on a system level? If not, which subsystems does this directly affect? Consider the interfaces as well. Which subsystems are responsible?
REQ.10.02		The communications subsystem shall obtain no less than two usable relay contacts per sol totaling at least 90 minutes of link time above a 10 degree elevation mask, and shall bound end-to-end latency from measurement to Earth receipt at no more than 26 hours.	7/28/2026	A single pass per sol leaves no recovery if geometry, occultation, or a fault costs the contact, so two are required and the 7 sols of onboard storage in REQ.03.04 absorbs the backlog when one is lost. The 26-hour bound is one sol of accumulation plus the 44-minute one-way light delay plus relay store-and-forward, which is the tightest figure that still puts a completed sol in front of Earth planners before the next one starts. 3 dB is the standard allowance for a proximity link whose orbit geometry, antenna pattern, and receiver noise floor are only coarsely known before flight, and at UHF the Jovian synchrotron contribution to system noise temperature is the hardest term to bound from the ground. A 1E-6 coded error rate keeps frame loss inside what the consensus protocol already tolerates at 30% packet loss (REQ.04.01). The beacon exists because the link budget shows a direct X-band link on an LGA closes at only ~7 bps at 4.2 AU, enough to report state but nowhere near enough to run the mission. That number is what makes the relay architecture non-optional rather than convenient.	ConOps minute-by-minute breakdown (Communication Relay Pass, 90 min/sol); REQ.03.04 seven-sol onboard storage; REQ.06.03 daily summary packet; NASA DSNE deep-space link design standards; Mars Relay Network pass scheduling heritage (MRO / Odyssey)	COMMs, CDH, Science/Payload, GNC
REQ.10.03		The rover-to-relay proximity link shall close with no less than 3 dB of margin over the Eb/N0 required for a post-decoding bit error rate of 1E-6 at the worst-case 588 km slant range in both directions, and the rover shall retain an X-band direct-to-Earth engineering beacon of no less than 5 bps to a DSN 34 m station as a relay-loss fallback.	7/28/2026		Link budget analysis (3.5); JPL DSN Telecommunications Link Design Handbook 810-005; CCSDS 131.0-B turbo/LDPC coding performance; JPL Electra-Lite UHF transceiver specification; REQ.04.01 degraded-mode consensus	COMMs, Power, Structures, CDH

		Status?	Associated Risks with	VERIFICATION & VALIDATION (V&V)	Testing Method	Verification Plan	Validation Plan	Notes For
OBJ = Objective, REQ = Requirement, Constraint. CSR = Number Hierarchy = 01.02.03 (01 = Objective Parent Category, 02 = First Child Requirement, 03 = Second Child Requirement... etc)		Accomplished or unaccomplished?	List the associated risk numbers.			Verification - Does it meet the specification need? Are we building our product right?	Validation - Does it meet the customer need? Did we build the right product?	What work did you do for this requirement this semester
REQ.01.01	Unaccomplished		Risk-01: Single-point autonomy failure with no fallback human override		Hardware-in-the-loop simulation. Run a 72-hour autonomous mission sequence with a simulated 44-min communication blackout, and verify zero unresolved command dependencies on Earth.	Verify the software autonomously executes all planned traverses, consensus rounds, and science data collected without any Earth uplink during the blackout window.	Demonstrate that the mission objectives are achieved across three independent HILL runs without Earth support	
REQ.01.02	Unaccomplished		RISK-02: Accumulated single event upsets degrade consensus algorithm convergence before mission completion.		Co-60 gamma irradiation of all radiation-sensitive components to 110 krad (10% margin). Monitor functional performance throughout exposure.	Verify all components pass functional tests at end of life dose with no loss of required functionality	Validate the integrated rover and relay nodes complete simulated mission operations after TID exposure.	
REQ.01.03	Unaccomplished		RISK-03: Simultaneous radiation-induced failure of multiple relay nodes isolating the surface rover.		Network fault simulation: simulate progressive relay node failures and verify consensus detects degenerated topology and triggers reconfiguration before dropping below 2 active nodes.	Verify that fault detection and reconfiguration logic activates within one consensus cycle period upon detection of a node loss event.	Validate that the surface rover completes a traverse plan using map data from a degraded 2-relay configuration.	
REQ.02.01	Unaccomplished		RISK-04: Ice surface texture homogeneity reducing visual odometry feature tracking quality.		Field test on analog terrain: command 50m traverse segments and compare onboard state estimate to ground truth GPSD position across 20 runs under varying terrain conditions.	verify that position error remains within +/-0.5 m across a minimum of 20 traverse segments.	Validate that the rover navigates to and stops within +/-0.5 m of a designated waypoint representing a candidate insertion site.	
REQ.03.01	Unaccomplished		RISK-05: Ice shell heterogeneity or brine inclusions creating radar clutter that degrades thickness estimates.		Laboratory radar range measurements against ice targets of known thickness using dielectric matches analogs, verify +/- 10% accuracy across the 1-30 km depth range.	Verify that the radar sounder produces subsurface profiles with +/-10% depth accuracy against calibration targets before rover integration.	Validate that the subsurface map produces during a field traverse correctly identifies presurveyed thin ice candidate sites within the required spatial resolution.	
REQ.04.01	Unaccomplished		RISK-06: Radiation induced burst errors exceeding 30% packet loss on multiple links simultaneously, stalling consensus.		Software-in-the-loop consensus simulation: Introduce 30% packet loss on each inter-agent link and measure convergence time across 100 test runs.	Verify the consensus convergence time remains <10 minutes across all 100 test runs with 30% packet loss.	Validate that the rover initiates a traverse using a map update produced by the consensus protocol under the degraded link test condition.	

	Status?	ASSOCIATED RISKS	Associated Risks with	VERIFICATION & VALIDATION (V&V)	Testing Method	Verification Plan	Validation Plan	Notes For
OBJ = Objective, REQ = Requirement, Constraint. CSR = Number <i>Hierarchy = 01.02.03 (01 = Objective Parent Category, 02 = First Child Requirement, 03 = Second Child Requirement... etc)</i>	Accomplished or unaccomplished?		List the associated risk numbers.			<i>Verification - Does it meet the specification need? Are we building our product right?</i>	<i>Validation - Does it meet the customer need? Did we build the right product?</i>	What work did you do for this requirement this semester
REQ.03.02	Unaccomplished		RISK-05: Ice shell heterogeneity creating radar clutter that degrades thickness estimates; RISK-04: Ice surface texture reducing visual odometry quality (slope-stop logic dependency)		Bench test antenna mounting fixture: tilt rover chassis to 5° increments and measure antenna boresight deviation using an inclinometer. Verify pointing error stays within ±5° across the full range of surface slopes up to 20°.	Verify that antenna boresight deviation does not exceed ±5° of nadir across all chassis tilt angles up to 20° slope	Validate that the rover correctly halts traversal and resumes sounding when slope threshold is exceeded during an analog terrain field test.	
REQ.03.03	Unaccomplished		RISK-02: Accumulated single event upsets degrade system performance; RISK-05: Degraded radar return amplitude from insufficient transmission power		Hardware-in-the-loop power budget simulation: command simultaneous drive and sounding operations and monitor RTG output margin across 50 test cycles. Verify drive shutdown command fires before each sounding pulse.	Verify that drive motors are commanded off prior to every sounding transmission and that RTG output margin remains positive across all test cycles.	Validate that the rover completes a full simulated traverse sol with correct stop-and-sound behavior and no power fault events.	
REQ.03.04	Unaccomplished		RISK-06: Radiation induced burst errors exceeding 30% packet loss stalling consensus; RISK-05: Ice shell heterogeneity generating excess CLUTTER traces inflating data volume		Software-in-the-loop storage simulation: generate 7 sols of synthetic radar trace data at full volume and verify onboard storage does not overflow. Apply compression algorithm and measure output ratio across 100 data sets.	Verify that onboard storage accommodates 7 sols of uncompressed data and that compression ratio meets or exceeds 2:1 across all 100 test data sets.	Validate that compressed science data downlinked through the inter-agent mesh under 30% packet loss conditions can be fully reconstructed without data loss on the receiving node.	
REQ.05.01	Unaccomplished		RISK-07: Thermal control failure during extended night-mode causing battery cell damage and mission termination		Thermal-Vacuum chamber test: Subject the integrated rover to Europa analog thermal cycling (90K surface, simulated RTG waste heat input) and monitor avionics and battery temperatures across a full 23 hour sol cycle including night-mode.	Verify that avionics and battery temperatures remain within required bounds across three consecutive sol cycles in the thermal vacuum chamber with no active ground commands.	Validate that the rover completes a simulated surface traverse sol following a full night-mode thermal cycle without any subsystem degradation or heater fault events.	

	Status?	ASSOCIATED RISKS	Associated Risks with	VERIFICATION & VALIDATION (V&V)	Testing Method	Verification Plan	Validation Plan	Notes For
OBJ = Objective, REQ = Requirement, Constraint. CSR = Number Hierarchy = 01.02.03 (01 = Objective Parent Category, 02 = First Child Requirement, 03 = Second Child Requirement... etc)	Accomplished or unaccomplished?		List the associated risk numbers.			Verification - Does it meet the specification need? Are we building our product right?	Validation - Does it meet the customer need? Did we build the right product?	What work did you do for this requirement this semester
REQ.05.02	Unaccomplished		RISK-02: Accumulated single event upsets degrade consensus algorithm convergence before mission completion.		Hardware-in-the-loop fault injection: Inject simulated single-event latchup signals into each processor core and measure time from latchup detection to nominal operation restoration across 50 test runs per core.	Verify that the recovery from the latchup injection occurs within 30 seconds across all 50 test runs for each core, and that each event is correctly logged to non-volatile memory with timestamp and code ID.	Validate that the rover autonomously continues a planned traverse sequence without human intervention following a latchup event injected mid-traverse during an analog mission simulation.	
REQ.05.03	Unaccomplished		RISK-02: Accumulated single event upsets degrade consensus algorithm convergence before mission completion. RISK-07: Thermal control failure during extended night-mode causing battery cell damage and mission termination		Hardware-in-the-Loop power budget simulation: Run a full synthetic sol sequence with simultaneous drive, sounding, heating, and consensus relay operations and monitor RTG output margin continuously across 550 test cycles, including injected anomaly events.	Verify that the total per-sol energy draw does not exceed 90% of RTG nominal output across all 50 test cycles, including cycles with injected anomaly events that trigger unschedules heater and relay draws.	Validate that the rover completes a full simulated traverse sol with one injected latch up and one slope-stop event without triggering a power fault or dropping below that 10% reserve margin.	
REQ.06.01	Unaccomplished		RISK-05: Ice shell heterogeneity creating radar clutter that degrades thickness estimates; RISK-04: Ice surface texture reducing visual odometry quality (slope-stop logic dependency)		Field test on analog terrain: command the rover to execute a 100m traverse with stop-and-sound events at <100m intervals and measure total traverse distance completed and number of sounding stops executed across 10 runs.	Verify that the rover completes >100m of traverse per simulated sol and executes a sounding stop at every <100m interval across all 10 test runs without a ground uplink command.	Validate that the subsurface map generated from a full simulated traverse sol contains valid radar traces at <100m along-track spacing covering the full traverse distance.	
REQ.06.02	Unaccomplished		RISK-05: Ice shell heterogeneity creating radar clutter that degrades thickness estimates;		Software-in-the-loop simulation: inject synthetic radar trace datasets containing pre-seed candidate sites at known positions and measure whether the onboard algorithm flags the correct site within one sol of the criteria being met across 50 test runs.	Verify that the site-selection algorithm correctly flags all pre-seeded candidate sites within one sol of criteria satisfaction and produces zero false positive flags on CLUTTER or NOISE_FLOOR traces across all 50 runs,	Validate that during an analog terrain field traverse, the algorithm flags a candidate site matching pre-surveyed thin-ice analog targets within one sol of the rover reaching the qualifying zone.	

	Status?	ASSOCIATED RISKS	Associated Risks with	VERIFICATION & VALIDATION (V&V)	Testing Method	Verification Plan	Validation Plan	Notes For
OBJ = Objective, REQ = Requirement, Constraint. CSR = Number Hierarchy = 01.02.03 (01 = Objective Parent Category, 02 = First Child Requirement, 03 = Second Child Requirement... etc)	Accomplished or unaccomplished?		List the associated risk numbers.			Verification - Does it meet the specification need? Are we building our product right?	Validation - Does it meet the customer need? Did we build the right product?	What work did you do for this requirement this semester
REQ.06.03	Unaccomplished		RISK-06: Radiation induced burst errors exceeding 30% packet loss stalling consensus; RISK-05: Ice shell heterogeneity creating radar clutter that degrades thickness estimates		Earth-based scientists and mission planners require timely sol-by-sol progress data to assess whether mission objectives are being met and to prepare contingency plans within the non-deterministic relay window. With a 44-minute light delay, the rover cannot wait for a downlink command; it must autonomously initiate transmission when the data is ready. The 2-hour bound ensures the daily summary reaches before the next sol's traverse planning begins, maintaining operational cadence. Traceable to OBJ.01 and OBJ.04	Verify that the daily summary packet is autonomously transmitted and confirmed received at the relay node within 2 hours of data collection completion across all 30 test runs, including degraded-link runs.	Validate that a complete daily summary record is successfully reconstructed from a compressed packet received through the inter-agent mesh under 30% packet loss conditions during an end-to-end mission simulation.	
REQ.07.01	Unaccomplished		RISK-08: Landing leg plastic deformation or buckling under 12 g vertical /6g lateral loads, causing frame misalignment and mobility failure.		FEA nonlinear analysis: Model landing leg assembly under peak 12g vertical and 6 g lateral loads; verify material yield safety factor >= 1.5 across all elements. Test on two candidate leg geometries (tapered strut vs constant diameter). Thermal-vacuum cycling test: Subject candidate fastener materials and deck plate samples to 30 cycles of simulated Europa surface (90K hold for 2hr, warm to -20C, cool back). Measure fracture toughness, dimensional change, and surface crack initiation via SEM.	Verify leg structure remains elastic (no permanent deformation) under 120% of worse-case landing load envelope. Confirm peak stress stays below material yield to all four leg attachment points across 50 simulated touchdown scenarios.	Validate by drop test: Lower fully integrated rover 1.2m onto rigid analog regolith target at 12 g equivalent impact; measure leg deformation, frame alignment post-impact, and mobility performance on 20 slopes.	
REQ.07.02	Unaccomplished		RISK-09: Brittle fracture of fasteners or primary structure during thermal cycling 90k -> operational range (-20C); Risk-10: Reduced fatigue margin under sustained cryogenic vibration.			Verify all fasteners and primary deck material remain in ductile regime (Charpy impact energy >5 J/cm^2) at 90K. Confirm zero crack initiation on structure after 30 thermal cycles. Measure dimensional stability: <2% shrinkage over full temperature range.	Verify all fasteners and primary deck material remain in ductile regime (Charpy impact energy >5 J/cm^2) at 90K. Confirm zero crack initiation on structure after 30 thermal cycles. Measure dimensional stability: <2% shrinkage over full temperature range.	

		Status?	Associated Risks with	VERIFICATION & VALIDATION (V&V)	Testing Method	Verification Plan	Validation Plan	Notes For
OBJ = Objective, REQ = Requirement, Constraint. CSR = Number Hierarchy = 01.02.03 (01 = Objective Parent Category, 02 = First Child Requirement, 03 = Second Child Requirement... etc)		Accomplished or unaccomplished?	List the associated risk numbers.			Verification - Does it meet the specification need? Are we building our product right?	Validation - Does it meet the customer need? Did we build the right product?	What work did you do for this requirement this semester
REQ.07.03	Unaccomplished		RISK-02: Accumulated single event upset if shielding dose rate exceeds 1 krad/sol threshold; Risk-11: Mass overflow if shielding vault exceeds 28.5 kg budget.		GEANT4 Monte Carlo radiation transport simulations: Model jovian particle spectrum incident on rover. Vary shielding layer (polyethylene, lead, or hybrid) and thickness; compute dose rate at flight computer location across 100,000 particle histories. Cross-validate with SRIM for heavy ion stopping power.	Verify shielding material and geometry reduce external dose rate (~5 krad/sol ambient) to <=1 krad/sol at flight computer board level. Confirm mass of vault assembly (enclosure + fasteners + thermal interface) stays <=3.2 kg allocated within structures budget (28.5kg total).	Validate by placing radiation-sensitive test diodes inside shielded vault prototype, exposing Co-60 gamma source at calibrated dose rate, and confirming internal diode reading matches GEANT4 prediction within +/-15%. Then validate via end-to-end HILL: run mission simulation with shielded flight computer after accumulated 110 krad exposure (10% margin) and verify CDH autonomy performance (latchup detection/recover per REQ. 05.02) remains nominal.	
REQ.08.01	Unaccomplished		RISK-12: RTG output degradation exceeding predicted 4.8%/yr curve, eroding surface ops energy margin		Electrical ground test of RTG simulator on flight representative bus: measure deliver power at simulated BOL and end-of-cruise degradation states across full thermal operating range.	Verify > 110 We at BOL conditions and >80 We at simulated 6-year degradation state, confirming >1.9 kWh/sol delivered to the bus.	Validate that a full simulated surface sol (traverse, sounding, relay pass, night mode) completes on <80 We continuous input plus battery load-leveling without a power fault.	
REQ.08.02	Unaccomplished		RISK-07: Thermal control failure during extended night-mode causing battery cell damage and mission termination		Thermal-vacuum endurance test replicating a 12-hour night-mode cycle at Europa analog temperatures with survival heater and avionics load profile applied.	Verify pack delivers >2.4 kWh usable and maintains bus voltage above cutoff through the full 12-hour night discharge with cells held above -20C	Validate the rover completes three consecutive simulated sol cycles (peak load-leveling by day, 12-hour survival discharge by night) with no capacity fault or cell degradation	
REQ.08.03	Unaccomplished		RISK-02: Accumulated single event upsets degrade consensus algorithm convergence before mission completion. RISK-13: PDU logic or sensor fault causing persistent loss of a non-faulted load bus\		Hardware-in-the-loop fault injection: Inject overcurrent and undervoltage events on each load bus and measure isolation time and critical-load continuity across a50 runs per bus.\	Verify fault isolation occurs within 100 ms in all injected cases, critical loads see no interruption, and every event is correctly 6 logged with timestamp and busID.	Validate the rover continues a planned traversal sol following a mid-sol injected bus fault, with autonomous recover and no loss of science data collection.	

		Status?	Associated Risks with	VERIFICATION & VALIDATION (V&V)	Testing Method	Verification Plan	Validation Plan	Notes For
OBJ = Objective, REQ = Requirement, Constraint. CSR = Number Hierarchy = 01.02.03 (01 = Objective Parent Category, 02 = First Child Requirement, 03 = Second Child Requirement... etc)		Accomplished or unaccomplished?	List the associated risk numbers.			Verification - Does it meet the specification need? Are we building our product right?	Validation - Does it meet the customer need? Did we build the right product?	What work did you do for this requirement this semester
REQ.09.01	Unaccomplished	ASSOCIATED RISKS	RISK-14: Avionics or radar electronics overheating during peak daytime operations from insufficient radiator rejection		Thermal-vacuum hot-case test: run full electronics load (radar transmit, drive, comms) with a 90 K chamber sink and simulated RTG waste-heat input; measure vault interior temperature at steady state.	Verify the vault interior stays below +40 C at peak dissipation across three consecutive simulated sols in the thermal-vacuum chamber.	Validate the rover completes a full simulated daytime traverse-and-sounding sol with no electronics thermal derating or fault events.	
REQ.09.02	Unaccomplished		RISK-15: Actuator lubricant stiffening or mechanism binding from cold-start motion below survival temperature		Cold-start bench test in TVAC: cold-soak actuators to 90 K, command warm-up, and measure time to reach -40 C plus first-motion torque margin across 10 runs.	Verify all actuators reach above -40 C within 60 minutes of warm-up initiation and produce nominal first-motion torque across all 10 cold-start runs.	Validate the rover executes a commanded traverse immediately following a full cold-soak and warm-up cycle with no mechanism fault.	
REQ.09.03	Unaccomplished		RISK-07: Thermal control failure during extended night-mode causing battery cell damage and mission termination		Thermal-vacuum survival test: instrument the MLI-wrapped enclosure at a 90 K sink with RTG waste-heat input; measure passive heat leak and the electric heater power required to hold survival temperatures over a 12-hour night.	Verify supplemental electric heater draw stays <=50 W while cells and avionics remain above survival minimums through the full 12-hour night discharge.	Validate the rover completes three consecutive simulated sols (daytime ops, 12-hour night survival) with the selected insulation architecture and no thermal fault.	
REQ.10.01	Unaccomplished		RISK-16: Per-sol data generation outpacing the relay return-link volume, forcing lossy science triage onboard		End-to-end throughput test: play a synthetic 7-sol data set through the flight radio into a relay emulator at the 128 kbps requirement floor and measure delivered volume per emulated pass.	Verify no less than 55 MB of compressed data is delivered per emulated sol at 128 kbps within the 90-minute relay window across 20 consecutive passes.	Validate that a full simulated sol of rover-generated data is reconstructed on the ground side with no loss of radar traces flagged NOMINAL.	

	Status?	ASSOCIATED RISKS	Associated Risks with	VERIFICATION & VALIDATION (V&V)	Testing Method	Verification Plan	Validation Plan	Notes For
OBJ = Objective, REQ = Requirement, Constraint. CSR = Number Hierarchy = 01.02.03 (01 = Objective Parent Category, 02 = First Child Requirement, 03 = Second Child Requirement... etc)	Accomplished or unaccomplished?		List the associated risk numbers.			Verification - Does it meet the specification need? Are we building our product right?	Validation - Does it meet the customer need? Did we build the right product?	What work did you do for this requirement this semester
REQ.10.02	Unaccomplished		RISK-17: Relay orbit geometry or Jupiter occultation collapsing the per-sol contact window below the volume needed to stay ahead of data generation		Orbit propagation and contact analysis: propagate both relay agents over the full 30-sol surface mission and count contacts and cumulative link time above the 10 degree mask, including Jupiter occultation intervals.	Verify no less than two contacts and 90 minutes of usable link time occur on every sol of the 30-sol mission, and that the worst-case sol still closes the volume requirement of REQ.10.01.	Validate that the rover carries a full sol of data forward and clears the backlog on the next contact when one pass is deliberately withheld during a simulated 3-sol operations run.	
REQ.10.03	Unaccomplished		RISK-18: Jovian synchrotron noise at UHF raising relay receiver system noise temperature above the assumed 3000 K and eroding return-link margin		Radiated compatibility test in an anechoic chamber with the flight helix on a rover mock-up, plus a bit error rate sweep against an injected noise floor scaled to the assumed system noise temperature.	Verify measured Eb/N0 at the worst-case 588 km slant range exceeds the coded 1E-6 threshold by no less than 3 dB in both directions, and that the X-band chain closes a 5 bps beacon at 4.2 AU.	Validate that the rover autonomously falls back to the X-band beacon and continues safe-mode operations when both relay links are withheld for a full simulated sol.	

OBJ = Objective,
REQ = Requirement, **CSR =**
Constraint. *Number*
Hierarchy = 01.02.03 (01 =
Objective Parent Category, 02 =
First Child Requirement, 03 =
Second Child Requirement... etc)

REQ.01.01

REQ.01.02

REQ.01.03

REQ.02.01

REQ.03.01

REQ.04.01

OBJ = Objective,
REQ = Requirement, **CSR =**
Constraint. *Number*
Hierarchy = 01.02.03 (01 =
Objective Parent Category, 02 =
First Child Requirement, 03 =
Second Child Requirement... etc)

REQ.03.02

REQ.03.03

REQ.03.04

REQ.05.01

OBJ = Objective,
REQ = Requirement, **CSR =**
Constraint. *Number*
Hierarchy = 01.02.03 (01 =
Objective Parent Category, 02 =
First Child Requirement, 03 =
Second Child Requirement... etc)

REQ.05.02

REQ.05.03

REQ.06.01

REQ.06.02

OBJ = Objective,
REQ = Requirement, **CSR =**
Constraint. *Number*
Hierarchy = 01.02.03 (01 =
Objective Parent Category, 02 =
First Child Requirement, 03 =
Second Child Requirement... etc)

REQ.07.03

REQ.08.01

REQ.08.02

REQ.08.03

OBJ = Objective,
REQ = Requirement, **CSR =**
Constraint. *Number*
Hierarchy = 01.02.03 (01 =
Objective Parent Category, 02 =
First Child Requirement, 03 =
Second Child Requirement... etc)

REQ.09.01

REQ.09.02

REQ.09.03

REQ.10.01

Timestamp_UTC	Sol	Pressure_mbar	AirTemp_C	GroundTemp_C	Humidity_pct	WindSpeed_mps	WindDir_deg	Dust_OpticalDepth	Radiation_Wm2	Data_Quality_Flag
---------------	-----	---------------	-----------	--------------	--------------	---------------	-------------	-------------------	---------------	-------------------

Time (UTC or Mission Elapsed Time)	Robot Function / Mode	Primary Objective / Milestone	Operation Mode (Teleop / Autonomous / Semi-Autonomous / Scripted)	Inputs (e.g., command, signal, state)	Expected Outputs (e.g., telemetry, sensor data)	Data Products Generated	Verification Method	Associated Risk or Sensitivity (High/Med/Low)	Dependencies / Predecessor Actions
Launch day, 0-1 (Mission Elapsed Time)	Launch And Commissioning	Spacecraft Separation and initial health checks	Scripted/Autonomous	Launch vehicle separation signal, onboard startup sequences	Spacecraft status telemetry, power system telemetry, communication link confirmation	Initial spacecraft health report, subsystem status logs	Verify successful separation communication establishment, and nominal subsystem health/	High	Successful vehicle loading and spacecraft deployment
Cruise Phase, Years 0-6	Cruise And Approach	Maintain spacecraft health and perform trajectory correction maneuvers	Semi-Autonomous	Ground-planned trajectory updates, star tracker measurements, navigation state estimates	Navigation telemetry, spacecraft health telemetry, maneuver confirmation	Trajectory correction logs, navigation solution history	Verify trajectory remains within mission tolerance and all spacecraft systems remain operational	Medium	Completion of launch and commission phase
Europa Orbital Insertion and Reconnaissance	Europa Orbit Insertion And Orbital Reconnaissance	Deploy orbital agents and identify suitable landing region	Autonomous	Orbital state estimates, terrain imaging data, radar survey measurements	Surface maps, hazard maps, candidate landing sites	Global terrain map, landing site assessment report	Verify successful orbital insertion, orbiter deployment, and identification of viable landing zones.	High	Successful arrival at Europa and orbital insertion burn
Surface Operations Day 0-5	Landing, Surface Checkout, And Site Characterization	Deploy rover, verify system health, and calibrate science instruments.	Autonomous	Landing state, rover deployment commands, sensor calibration routines	Health telemetry, localization estimates, calibration measurements	System checkout report, calibration datasets, local terrain map	Verify successful rover deployment, communications, localization accuracy, and payload calibration	High	Reconnaissance complete and landing site selected
Surface Operations Day 5-35	Surface Survey And Ice Characterization	Map subsurface ice structure and identify candidate cryobot insertion sites	Autonomous	Ground-penetrating radar returns, orbital terrain map, localization state estimations	Ice thickness estimates, terrain classifications, rover position estimates	Subsurface radar maps, ice thickness profiles, candidate cryobot site database, consensus terrain maps	Verify radar measurements meet required accuracy and candidate sites satisfy mission selection criteria	Medium	Successful rover deployment and payload calibration
Surface Operations Day 35-50	Data Transmission And End Of Mission	Downlink mission data, select final cryobot site, and safely passivate system	Semi-Autonomous	Stored science data, final site selection results, mission completion commands	Mission archive transmission, final health telemetry, passivation confirmation	Final mission report, validated cryobot insertion site package, complete science archive	Verify all science data successfully transmitted and spacecraft enters safe passivated state	Low	Completion of science objectives and final site selection

Estimated Duration	Notes
-----------------------	-------

1 day

~6 years

30 Days

5 Days

30 Days

15 days

Board	Component	Part name/Notes	Power Usage - On (W)	End efficiency	Heat Generation - On (W)	Minimum Survival Temperature	Minimum Operating Temperature	Maximum Operating Temperature	Maximum Survival Temperature
Payload	GPR Radar Sounder	REASON-heritage dual-freq sounder	45	0.95	2.25	-40	-30	50	60
EPS	Battery Packs	Li-ion, 2400 Wh (12 h survival)	0	1	0	-40	-20	40	50
EPS	PDU	PCDU, MMRTG EPS heritage	12	0.93	0.84	-30	-20	50	60
Mobility	Drive Motors (4x)	Cryo-rated brushless, MSL heritage	84	0.9	8.4	-40	-30	50	60
GNC	Flight Computer (OBC)	RAD750-class, vault-mounted	18	0.95	0.9	-30	-20	50	60
GNC	I/O & Storage Boards	Logger + ADC stack	6	0.95	0.3	-30	-20	50	60
GNC	Nav Cameras	Stereo VO pair, EECAM heritage	4	0.95	0.2	-40	-30	50	60
Thermal	Heaters	KHLVA-type film heaters	40	1	0				
Thermal	Thermal Sensors	ADT7311-class	0.5	0.95	0.025	-50	-40	85	125
Comms	UHF Radio	Electra-Lite heritage, helix	35	0.95	1.75	-30	-20	50	60
Comms	X-band Transponder	SDST + LGA	45	0.95	2.25	-30	-20	50	60
Total			289.5	0.942	16.91				

Each criterion was evaluated against every other using a pairwise comparison method. Values were assigned in 0.25 increments between 0 and 1, where:

- A value of 0.5 indicates equal importance.
- Values of 0.25 or 0.75 indicate one criterion is moderately more important than the other.
- Values of 0 or 1 reflect extreme preference toward one criterion, rendering the other negligible.

The comparisons are symmetric such that the sum of each pair equals 1. Total weights were summed and normalized to create priority values for each criterion.

Criteria	Mass	Power Demand	Complexity/	Radiation Durability	Reliability/TRL	Weight	Normalized
Mass		0.25	0.5	0.25	0.25	1.25	0.13
Power Demand	0.75		0.75	0.5	0.5	2.5	0.25
Complexity/ Integration	0.5	0.25		0.25	0.25	1.25	0.13
Radiation Durability	0.75	0.5	0.75		0.5	2.5	0.25
Reliability/TRL	0.75	0.5	0.75	0.5		2.5	0.25
Total						10	

	Mass	Power Demand	Complexity/	Radiation Durability	Reliability/TRL
5	< 5 kg added	< 25 W	Simple installation,	Flight-proven; negligible	TRL 8-9
4	5-9 kg	25-50 W	Moderate integration	Minor degradation;	TRL 6-7
3	10-14 kg	50-100 W	Custom mounts	Some degradation;	TRL 5-6
2	15-20 kg	100-150 W	Multiple interfaces /	Noticeable degradation /	TRL 4-5
1	> 20 kg	> 150 W	New mechanisms /	Strongly affected /	TRL < 3

Analysis of Night Survival Thermal Architecture							
Decision Criteria		MLI and heaters		Aerogel and heaters		PCM with heat pipes	
Criteria	Weight	Score	Weighted Score	Score	Weighted Score	Score	Weighted Score
Mass	0.13	4	0.52	4	0.52	2	0.26
Power Demand	0.25	5	1.25	3	0.75	4	1.00
Complexity/ Integration	0.13	4	0.52	3	0.39	2	0.26
Radiation Durability	0.25	5	1.25	3	0.75	3	0.75
Reliability/TRL	0.25	5	1.25	3	0.75	2	0.50
Total Score		4.79		3.16		2.77	

Each criterion was evaluated against every other using a pairwise comparison method. Values were assigned in 0.25 increments between 0 and 1, where:

- A value of 0.5 indicates equal importance.
- Values of 0.25 or 0.75 indicate one criterion is moderately more important than the other.
- Values of 0 or 1 reflect extreme preference toward one criterion, rendering the other negligible.

The comparisons are symmetric such that the sum of each pair equals 1. Total weights were summed and normalized to create priority values for each criterion.

Criteria	Mass	Power Demand	Data Rate	Pointing & Ops	Reliability/TRL	Weight	Normalized
Mass		0.25	0.25	0.5	0.25	1.25	0.13
Power Demand	0.75		0.5	0.75	0.5	2.5	0.25
Data Rate Capability	0.75	0.5		0.75	0.5	2.5	0.25
Pointing & Ops Complexity	0.5	0.25	0.25		0.25	1.25	0.13
Reliability/TRL	0.75	0.5	0.5	0.75		2.5	0.25
Total						10	

	Mass	Power Demand	Data Rate	Pointing & Ops	Reliability/TRL
5	< 3 kg	< 25 W	> 2 Mbps	Hemispherical; no	TRL 8-9; flight-proven
4	3-5 kg	25-50 W	512 kbps - 2 Mbps	Fixed; ±30° usable beam	TRL 7; minor
3	5-8 kg	50-75 W	128-512 kbps	Body-pointed or single-	TRL 5-6; unqualified for
2	8-12 kg	75-100 W	32-128 kbps	Coarse two-axis pointing	TRL 4-5; prototype only
1	> 12 kg	> 100 W	< 32 kbps	Closed-loop tracked	TRL < 3

Analysis of Rover-to-Relay Proximity Link Configuration							
Decision Criteria		UHF quadrifilar helix (401.6 / 437.1 MHz)		S-band patch array (2.2 GHz)		X-band steerable HGA cross-link (8.4 GHz)	
Criteria	Weight	Score	Weighted Score	Score	Weighted Score	Score	Weighted Score
Mass	0.13	5	0.65	4	0.52	2	0.26
Power Demand	0.25	4	1.00	3	0.75	2	0.50
Data Rate Capability	0.25	4	1.00	5	1.25	5	1.25
Pointing & Ops Complexity	0.13	5	0.65	4	0.52	1	0.13
Reliability/TRL	0.25	5	1.25	3	0.75	2	0.50
Total Score		4.55		3.79		2.64	

The UHF quadrifilar helix scores 4.55 against 3.79 for the S-band patch array and 2.64 for the X-band gimbaled cross-link. The X-band option has the raw throughput but fails the other two criteria: a two-axis gimbal is a moving mechanism that has to survive 90 K and 100 krad, and it needs closed-loop tracking of a relay the rover cannot see coming. S-band buys data rate the mission does not need, since the link budget shows the UHF return already closes at 128 kbps with 18.3 dB of margin and is bounded by the transceiver's 2048 kbps ceiling rather than by RF. The helix is also already in the mass budget at 0.86 kg and carries TRL 9 through Electra and Electra-Lite on exactly this kind of proximity link. Selected: UHF quadrifilar helix.